A Comparative Study of Audio Feature Representations for Environmental Sound Classification

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Research question

How do MFCC and Log-Mel feature representations compare in accuracy, macro-F1 and computational cost on ESC-50 when evaluated using the same RBF-SVM pipeline?

Working hypothesis: Log-Mel will achieve higher macro-F1, while MFCC will offer lower dimensionality and faster processing.

Background and motivation

Environmental sound classification identifies everyday acoustic events for context-aware devices, monitoring and assistive systems. These sounds range from tonal and stable to transient and noisy, so performance depends on the information retained by the feature representation. MFCC compactly summarises the spectral envelope on a perceptual scale but may discard local detail. Log-Mel spectrograms preserve Mel-band energy more directly, potentially improving discrimination at greater dimensional and computational cost. A controlled comparison using the same classifier and evaluation protocol is therefore needed to determine whether this richer representation offers a practical benefit.

Literature review and research gap

Piczak introduced ESC-50 as a balanced dataset of 2,000 five-second recordings across 50 classes, together with source-aware predefined folds [1]. The best handcrafted-feature baseline, a random forest classifier using MFCC and zero-crossing-rate features, achieved 44.3% accuracy, demonstrating both the difficulty of the dataset and the usefulness of a reproducible benchmark. Chu et al. showed that combining MFCC with Matching-Pursuit time-frequency features improved recognition on a different environmental sound dataset [2]. This supports the idea that complementary temporal and spectral information can be useful, but it does not establish that one representation is superior on ESC-50.

Piczak later obtained 64.5% accuracy on ESC-50 using a convolutional neural network with Log-Mel spectrograms, deltas, augmentation and segment-level voting [3]. The result demonstrates that Log-Mel is a feasible representation, but its effect is confounded with changes in the classifier and training procedure. Gourisaria et al. recently compared MFCC- and STFT-based systems [4], although the two features were applied to different datasets, preventing a direct representation-level conclusion. The broader survey by Chachada and Kuo also shows that reported performance depends jointly on features, dimensionality reduction and classifier choice [5].

Across these studies, no reviewed experiment performs the proposed controlled comparison: MFCC and Log-Mel extracted from the same ESC-50 clips, evaluated with the same RBF-SVM, official folds, tuning policy, predictive metrics and computational measurements. This project addresses that gap by changing the representation while controlling the remaining pipeline.

Proposed methodology

The independent variable will be audio feature representation: MFCC or Log-Mel. The complete ESC-50 dataset will be used with its five official source-aware folds. Each audio clip will be loaded as mono at 44,100 Hz and verified as five seconds long. No data augmentation will be used. Both representations will share a Hann window, an FFT and window length of 2,048 samples, a hop length of 512 samples and a 128-band Mel filter bank. Mel power will be converted to decibels using a per-clip maximum reference and an 80 dB floor.

The Log-Mel representation will retain all 128 bands. The MFCC representation will retain 40 coefficients computed from the same Log-Mel matrix. Each feature trajectory will be summarised across time using its mean and standard deviation, producing fixed-length vectors of 256 values for Log-Mel and 80 values for MFCC. This common pooling rule allows both inputs to be evaluated by the same classifier while keeping the feature construction transparent.

Both representations will be classified with an RBF-SVM inside a scikit-learn Pipeline containing StandardScaler. Evaluation will use nested official-fold validation. In each outer iteration, one official fold will be reserved for testing. Hyperparameters will be selected only from the other four folds using C values of 0.1, 1, 10 and 100, and gamma values of scale, 0.001, 0.01 and 0.1. Scaling will be fitted only within each training split. Mean inner macro-F1 will select the model before it is refitted on all outer-training data and evaluated once on the untouched test fold. This procedure prevents test-data leakage and gives every clip exactly one out-of-fold prediction.

Evaluation and expected outcomes

The primary outcomes will be accuracy and macro-F1, reported for each fold and as mean plus standard deviation across folds. All out-of-fold predictions will be combined to construct a row-normalised 50-class confusion matrix and identify categories that benefit from either representation. Feature-extraction time, final model-training time and inference time per clip will also be measured on the same computer. The expected result is not necessarily that Log-Mel always performs better. Instead, the experiment will quantify whether any predictive improvement justifies its larger representation and computational cost.

Feasibility and reproducibility

The study is feasible with Python, Jupyter, librosa and scikit-learn and does not require GPU training. Work will proceed through dataset validation, shared feature extraction, a small smoke test, the complete nested five-fold experiment, and final analysis. Parameters will be stored in a configuration file, package versions will be recorded, and random seed 42 will be used where applicable. Raw audio will remain outside GitHub, while code, setup instructions, configurations, predictions, metrics and figures will be version controlled.

Expected contribution and limitations

The contribution will be a transparent performance-efficiency comparison rather than a new state-of-the-art model. Conclusions will be limited to ESC-50, the selected RBF-SVM and the frozen preprocessing settings. Log-Mel also produces a larger vector than MFCC, and time pooling removes temporal order. These limitations will be reported explicitly so that the results are interpreted as evidence about the two defined pipelines rather than universal claims about all audio classifiers.

Weeks 1–13 timeline

Weeks	Planned work
1–2	Topic selection and preliminary literature review
3	Finalise proposal and GitHub structure
4–5	Download and validate ESC-50; implement shared preprocessing
6–7	Extract MFCC and Log-Mel features
8	Conduct smoke tests and verify the evaluation pipeline
9–10	Run nested five-fold experiments
11	Analyse metrics, confusion matrices and computational cost
12	Complete documentation, GitHub Pages and reproducibility checks
13	Finalise report, code and project demonstration

Project repository

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GitHub Username: SKY-XHL

Repository: ELEC5305 project GitHub repository

Project Website: https://github.com/SKY-XHL/elec5305-project-500051742

References

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